COMP 3981: Project Description (Part #1)

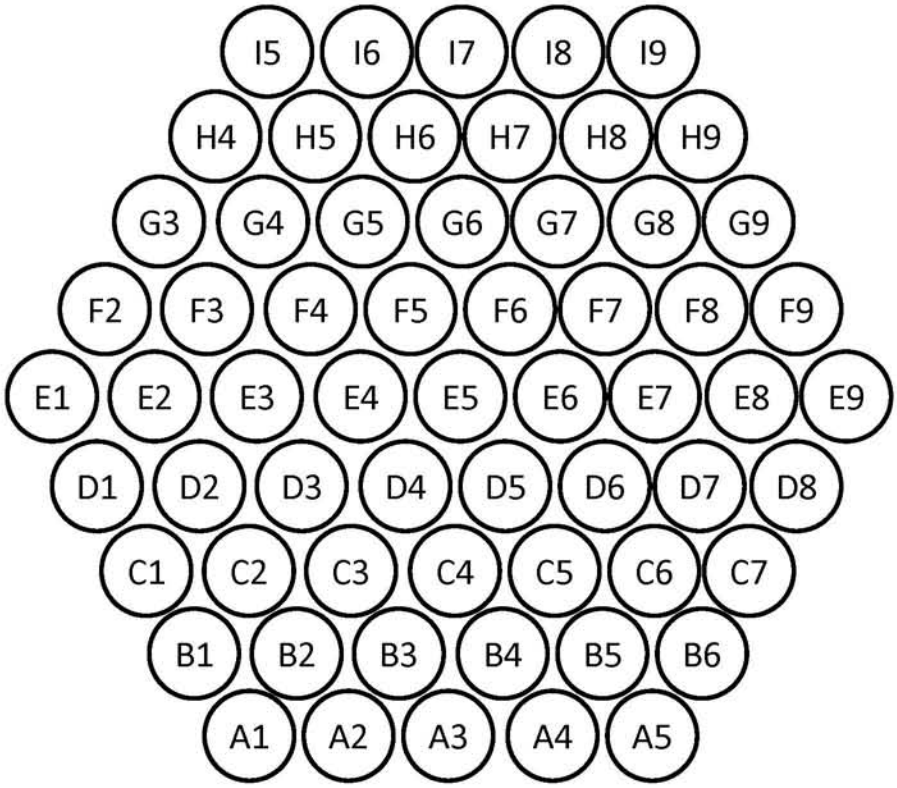
Problem Formulation and Graphical User Interface (GUI) Implementation

General Instruction:

For this part of the project, you will work with your team members on the game board representation, move notation, problem formulation as well as design and implement the graphical user interface for the abalone game-playing agent.

I Game Board Representation

For the abalone game-playing agent, you may use any game board representation (coordinate system) within your program, however, the input and output interfaces must conform to the game board coordinate system depicted below:



II Move Notation

You will also define the notation(s) to represent all possible moves of the marbles. Ensure that your notation(s) take into account:

- moves of a single marble
- in-line moves of two or three marbles
- side-step moves of two or three marbles

and provide example moves to illustrate your defined notation(s). (**Note:** You do not need to consider the legality of the resulting move at this time, i.e., assume that the surrounding circular spaces are empty and that the marbles are away from the edge of the board.)

III Problem Formulation

The problem formulation should include the following components:

a. State Representation
b. Initial State
c. Actions
d. Transition Model
e. Goal Test

IV GUI Design and Implementation

The GUI design and implementation (in any language) should include, at the minimum, the following components defined in (a), (b) and (c):

a. Input
   i. Game: start, stop, pause, reset, undo last move
   ii. Move input per player: drag and drop or text box
b. Output
   i. Display game score
   ii. Display moves taken per player
   iii. Display time taken per move by abalone game-playing agent (with complete history and total aggregate time)
   iv. Display next move as determined by the search strategy (with complete history)
   v. Update board layout after each player's move
c. Configuration
   i. Initial board layout: standard, german daisy and belgian daisy starting positions
   ii. Player color selection
   iii. Game mode: human vs. computer
   iv. Set per player, per-game move limit (same for both players)
   v. Set per player, per-move time limit (same or different for both players)
d. Additional features (as determined by each team)

Deliverables:

All work submitted is subject to the standards of conduct as specified in BCIT Policy 5104. For all deliverables, late submissions will not be accepted and you are allowed only one submission.

The deliverables, one per group, for Part #1 (both BCIT Learning Hub and in class submissions) are as follows:

1) [40 marks] [Mar 7, 2024 @1330] Submit your Part #1 report which includes the following sections:
   a. [10 marks] Game Board Representation and Move Notation (with example moves)
   b. [20 marks] Problem Formulation (State Representation, Initial State, Actions, Transition Model and Goal Test)
   c. [10 marks] Team Member Contribution

   in ***part1report_groupnumber.pdf***, to BCIT Learning Hub (Project Submission | Part 1: Report).

2) [20 marks] [Mar 7, 2024 @1330] Submit your source code and executable including detailed instructions on how to build and run the executable, in ***part1sourcecode_groupnumber.zip***, to BCIT Learning Hub (Project Submission | Part 1: Source Code). You are also required to indent and comment your code based on best practices - poorly indented and commented code will receive deductions, as appropriate. (**Note:** You will receive a mark of zero (i) if your code does not compile or execute on the (freshly-imaged) instructor terminal in the option lab (SE12-306) or (ii) you do not provide detailed instructions on how to (build and) run the executable.)

3) [30 marks] [Mar 7, 2024 @1330] GUI demonstration (in class)

Upcoming Project Due Dates:

Mar 21, 2024:	Project Part #2 - State Space Generation
Mar 28, 2024:	Project Part #3 - Search Strategy and Optimization
Apr 4, 2024,	
Apr 11, 2024:	Project Part #4 - Tournament and Final Deliverable